

Chongyang Xu (许重阳)

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Available for long-term internship (6-24 months) on a full-time basis (5+days/week) +86-13096391018 (same on WeChat)

EDUCATION

Sichuan University, Chengdu

Ph.D. student in Computer Science and Technology

B.Eng. in Software Engineering

Advisor: Prof. Ziliang Feng

Sept. 2023 – Present (Expected Graduation: Jun. 2028)

Sept. 2019 – Jun. 2023

Research Interests: research focuses on Vision-Language-Action (VLA), 3D pre-trained foundation models as world model, and dual-arm collaborative manipulation algorithms, with previous work in human motion capture and crowd analysis.

INTERNSHIP EXPERIENCE

Dexmal (原力灵机) Research Intern, mentored by Shuaicheng Liu

Mar. 2025 – Present (until early Feb. 2026)

- Reproduced state-of-the-art manipulation algorithms, ranging from **vision-based policies** (ACT, DP, DP3) to **VLA-based approaches**, including π_0 , RDT, and **X-VLA**, and successfully applied this expertise to secure **1st Prize** in both **Simulation** and **Real-world** tracks of the **RoboTwin Challenge** (CVPR 2025);
- Conducted research on bimanual manipulation policies with 3D representations, focusing on improving coordination and spatial reasoning in dual-arm manipulation; Explored the use of **2D pretrained models** (e.g., DINOv2, Stable Diffusion) to construct **hierarchical semantic fields** that enhance 3D point-cloud-based policies (**1 paper submitted to ICRA26**). Leveraging **3D pretrained foundation models** (e.g., VGGT, π^3) as **world models**, exploiting their latent geometric priors to improve perception, planning, and policy learning (**1 paper submitted to CVPR26**).
- Collected real-world manipulation data via **teleoperation** and reproduced **real-robot demos** on physical platforms, with hands-on experience debugging multiple robotic systems, including **Cobot Magic** and **DOS-W1** (both ALOHA);
- Organized **X-VLA evaluations on the real-robot benchmark RoboChallenge**, supporting both single- and dual-arm configurations across **ALOHA**, **Franka**, **UR5**, and **ARX5** platforms. Demonstrated expertise in joint-space and pose control, and performed **large-scale distributed hyperparameter tuning and optimization** for X-VLA on 48 NVIDIA H20 GPUs, achieving **real-robot success rates in which a 0.9B-parameter X-VLA model surpassed the performance of π_0 (3B parameters)**, conducted an in-depth investigation of **X-VLA's foundation model pre-training data and resource costs** (**my primary research interest for the upcoming internship**).

MEGVII (旷视科技) Research Intern, mentored by Shuaicheng Liu

Sept. 2024 – Mar. 2025

- Developed a high-fidelity simulation pipeline by integrating **PyBullet** physics with **Blender** rendering;
- Investigated advanced simulation of **deformable objects** (e.g., fluids, soft bodies, and cloth) within Blender's physics engine, creating various complex interaction demos to enhance the diversity and fidelity;
- Investigated multimodal alignment and representation learning for contact-rich manipulation tasks, and established a pipeline combining **Imitation Learning (IL)** and **Reinforcement Learning (RL)** to improve policy generalization; validated the approach by securing **1st Place** in the international ManiSkill-ViTac Challenge (CVPR 2025).

COMPETITIONS

1st Place - ManiSkill-ViTac Challenge Track 1

6th Annual Embodied AI Workshop, CVPR 2025

Team Members: Yixian Zou, Qinglun Zhang, **Chongyang Xu**, Shen Cheng, Haoqiang Fan, Shuaicheng Liu

Awarded to Team Dexmal-Bot; developed a unified representation by fusing and aligning **multimodal visuotactile information** within shared latent space, and then leverages **imitation learning with GT target supervision** for expert data collection, following by **reinforcement learning** to achieve closed-loop control based **solely on visuotactile feedback**.

1st Place - ManiSkill-ViTac Challenge Track 3

6th Annual Embodied AI Workshop, CVPR 2025

Team Members: **Chongyang Xu**, Yixian Zou, Qinglun Zhang, Shen Cheng, Haoqiang Fan, Shuaicheng Liu

Focused on optimizing **tactile sensor surface geometry and marker distributions** to maximize **signal discriminability**, enabling the downstream policy to perceive subtle contact interactions with higher fidelity at hardware-level.

1st Prize - RoboTwin Collaboration Challenge, Simulation Round 1

2nd MEIS Workshop, CVPR 2025

Team Members: Yixian Zou, **Chongyang Xu**, Qinglun Zhang, Shen Cheng, Ping Tan, Haoqiang Fan, Shuaicheng Liu

Focused on dual-arm VLA models (π_0 , RDT) for imitation learning in **simulated single-task** bimanual collaboration.

2nd Prize - RoboTwin Collaboration Challenge, Simulation Round 2

2nd MEIS Workshop, CVPR 2025

Team Members: Yixian Zou, **Chongyang Xu**, Qinglun Zhang, Shen Cheng, Ping Tan, Haoqiang Fan, Shuaicheng Liu

We further develop an **All-in-One** VLA model for generalization in simulated environments to handle **multiple tasks**.

Team Members: Yixian Zou, **Chongyang Xu**, Qinglun Zhang, Shen Cheng, Ping Tan, Haoqiang Fan, Shuaicheng Liu
Focused on dual-arm VLA models (π_0 , RDT) for **real-world bimanual collaboration**, including data collection and processing, intention abstraction, deployment with communication, and enhancing robustness in real environments.

PUBLICATIONS

Conference Publications

3. **Multi-Scale Hypergraph Relational Reasoning for Weakly Supervised Recognition of Group Activities.**
Chongyang Xu, Runtian Zheng, Ziliang Feng, Chengfang Zhang.
IEEE International Conference on Multimedia and Expo (ICME), 2025. **(Oral presentation)**
2. **Reconstructing Close Human Interaction with Appearance and Proxemics Reasoning.**
Buzhen Huang, Chen Li, **Chongyang Xu**, Dongyue Lu, Jinnan Chen, Yangang Wang, Gim Hee Lee.
Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR), 2025, pp. 17475–17485.
1. **Closely Interactive Human Reconstruction with Proxemics and Physics-Guided Adaption.**
Buzhen Huang, Chen Li, **Chongyang Xu**, Liang Pan, Yangang Wang, Gim Hee Lee.
Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR), 2024, pp. 1011–1021.

Selected Preprints & Under Review

3. **Action-Geometry Prediction with 3D Geometric Prior for Bimanual Manipulation.**
Chongyang Xu, Haipeng Li, Cheng Shen, Haoqiang Fan, Ziliang Feng, Shuaicheng Liu.
Submitted to CVPR 2026, Nov. 2025. 3D Foundation Models as World Model for Coordinated Bimanual Manipulation.
By building upon pre-trained 3D geometric foundation models like VGGT or π^3 , this work unifies geometry-aware latents with semantic features to jointly predict future action sequences and 3D scene evolutions, enabling robust, coordination-aware bimanual manipulation directly from RGB observations.
2. **Hierarchical Semantic Conditioning for Pose-aware Object Manipulation.**
Chongyang Xu, Cheng Shen, Haipeng Li, Haoqiang Fan, Ziliang Feng, Shuaicheng Liu.
Submitted to ICRA 2026, Sep. 2025. 3D Semantic Representation for Part-Aware Object Manipulation.
By lifting DINOv2 and Stable Diffusion features into hierarchical 3D semantic fields, this work explicitly models interactions between object parts to enable fine-grained understanding in diffusion-based policies, overcoming the semantic limitations of purely geometric representations for pose-aware manipulation.
1. **Reconstructing Multi-Person Interactions via Semantic-Geometric Graph Optimization.**
Chongyang Xu, Buzhen Huang, Wentao Tang, Ziliang Feng, Kun Li, Yangang Wang.
Submitted to CVPR 2026, Nov. 2025.

Technical Reports

2. **RoboChallenge: Large-scale Real-robot Evaluation of Embodied Policies** **First Real-world Benchmark**
Adina Yakefu, Bin Xie, **Chongyang Xu**, Enwen Zhang, Erjin Zhou, et al.
arXiv preprint arXiv:2510.17950, 2025.
1. **Benchmarking generalizable bimanual manipulation: Robotwin dual-arm collaboration challenge at cvpr 2025 meis workshop**
arXiv preprint arXiv:2506.23351, 2025.

RESEARCH PROJECTS

Evaluation of X-VLA for RoboChallenge Benchmarking

November 2025 - Present

- Reproduced and analyzed X-VLA pretraining and fine-tuning procedures to understand model behavior;
- Conducted large-scale fine-tuning and evaluated the resulting policies on the RoboChallenge benchmark;
- Focused on benchmarking and performance assessment of X-VLA for real-robot embodied tasks.

Exploration of Embodied Simulation in Blender

September 2024 - December 2024

- Developed and experimented with embodied simulation pipelines in Blender, controlling robotic arm motion;
- Integrated PyBullet with Blender's built-in physics engine to simulate rigid bodies, soft bodies, cloth, and fluid interactions;
- Conducted rendering and physics experiments to evaluate realistic interactions of deformable objects, enhancing the fidelity of simulated robotic manipulation scenarios.